Deadline-Aware Wireless Channel Selection

Partial feedback, packet deadlines, and changing interference

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https://github.com/Ayush-Singh-as/DeadlineHop

1. Problem and research grounding

A wireless sensor link must deliver queued packets before their deadlines while interference changes. Staying on a poor channel wastes service opportunities; hopping consumes time; transmissions can reveal the channel to a delayed follower. These coupled future effects motivate reinforcement learning under partial observation, without implying that conventional policies cannot work.

Dynamic multichannel access with selected-channel feedback motivates history-based DQN [1]. Radar anti-jamming research motivates temporal context [2], while hidden anti-jamming motivates action-dependent interference [3]. Their observation and reward models differ from ours. We implement an original simulator and Double DQN [4], rather than reproducing their scores or copying their code.

2. Environment and POMDP

One coordinated transmitter/receiver pair uses eight channels and an eight-packet queue. Packets arrive in 500 slots with Bernoulli probability sampled from 0.35-0.65; deadlines are 4, 8, or 12 slots. A 12-slot drain completes a 512-slot mission. Earliest-deadline-first service is fixed across every policy. Overflow rejects new packets; expired packets are counted explicitly.

The hidden state contains interference mode, masks, phase, timers, and pending follower responses. The 48-value observation contains tuned channel (8), queue presence/age/slack (24), clocks (2), previous action (9), previous outcome (4), and validity (1). H=16 stacks observations into 768 inputs; H=1 uses 48 and still sees the previous event. No mask, mode, or future arrival is provided to the policy.

Actions 0-7 choose a channel; action 8 waits. Changing channel occupies one slot without transmitting. Staying attempts one queued packet. Clean success probability is 0.98; interference multiplies it by 0.02. A packet arriving at a with deadline D can be served through a+D-1. Every transition checks generated = delivered + expired + overflow + queued.

3. Dynamics and learning objective

Training equally mixes a correlated sweep, ON/OFF bursts, a delayed follower, and episodes switching between these families. Sweeps block 2-4 adjacent channels and move with probability 0.25-0.75. Bursts block 2-4 channels. The follower detects emissions with probability 0.6-1, responds after 1-3 slots, and occupies its target for 2-5 slots. Current interference is fixed before the action; mode changes reset old pending responses.

For delivery s, expiry x, overflow b, retune h, and attempt u, the reward is:
r = s - 2(x+b) - 0.02h - 0.01(u+0.2h).
Gamma is 0.99. WAIT has no bonus. Costs use normalized units; physical watts or milliseconds are not claimed. The terminal mission clock is observed, and terminal targets do not bootstrap.

Two 128-unit ReLU layers output nine Q values. Double DQN selects the next action using the online network and evaluates it using the target network. Training uses Adam (0.0003), Huber loss, batch 64, replay 50,000, 5,000-step warm-up, one update per four steps, and target copies every 1,000 updates. Epsilon reaches 0.05 at 100,000 steps. Histories never cross episode or overwritten replay boundaries.

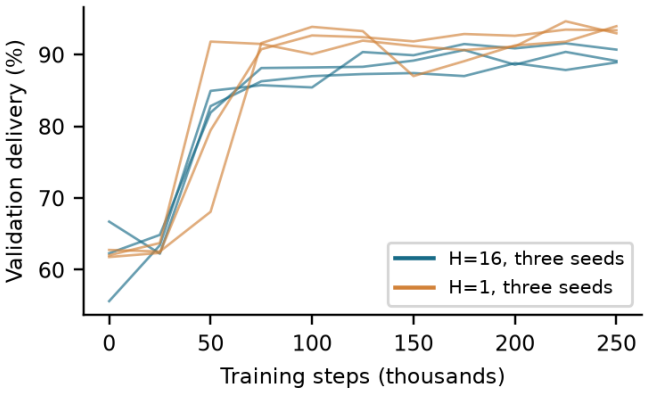


Figure 1. Validation delivery for all six independently trained runs. Each history variant receives 250,000 steps per seed. These curves are not final test performance.

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4. Evaluation protocol

We train seeds 11, 22, and 33 for each history length. Eighty fixed validation episodes select each seed's checkpoint by family-macro-average on-time delivery, with energy as a tie-breaker. The test manifest is frozen before training: 200 structured episodes, 100 no-interference/IID controls, and 100 stress episodes. Network weights stay frozen in evaluation; adaptive baselines update observed counts within each episode.

Baselines are fixed channel, random/cyclic hopping, stay-on-success/switch-on-failure, sliding-window UCB, and a history-conditioned myopic predictor. Hopping commits through retuning to a transmission. Fixed channel and UCB settings use validation only. The predictor fits smoothed context counts on 60 training episodes per candidate context length and selects length on validation. All policies use identical ACK feedback and EDF scheduling.

Paired scenarios share arrivals and pre-indexed random draws; reactive interference still responds to each policy's own actions. Delivery includes every generated packet, including overflow. We average episodes within families, then families equally within each trained seed. The table reports mean and seed range; three seeds do not support precise claims of statistical significance.

Policy	Delivery %	Seed range
Fixed channel	65.17	-
Random hopping	78.71	-
Cyclic hopping	76.76	-
Stay/switch	87.72	-
Window UCB	77.55	-
History predictor	94.56	-
Double DQN H=1	94.06	93.6-94.5
Double DQN H=16	90.80	89.6-92.2

Table 1. Held-out structured delivery; baseline dashes mean one scenario-indexed evaluation stream, not zero uncertainty. All seeds are retained.

5. Results and limitations

H=16 achieves **90.80%** structured delivery and H=1 achieves **94.06%**. The strongest conventional policy is History predictor at **94.56%**. H=16 trails it by 3.75 percentage points. The H=16 minus H=1 difference is -3.25 points. These measured results do not establish that longer history is necessary or that RL is the only effective approach.

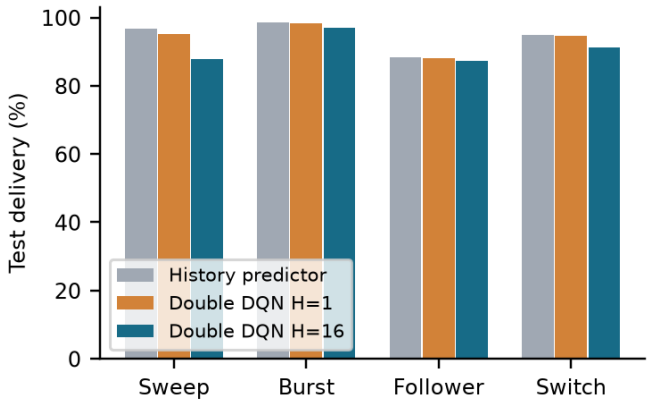


Figure 2. Mean held-out delivery by structured family; full baseline, control, stress, and seed tables are in the repository.

Normalized energy per delivery is 1.42 for H=16 and 1.37 for History predictor. H=16 delivery under no interference is 99.9% and under IID interference is 69.4%. IID checks validate the per-attempt analytical mean; they do not require all controllers to have equal packet delivery under retuning costs.

Stress tests separately increase load to 0.80, shorten deadlines to {2,4,6}, extend follower delay to 4, or accelerate sweeps to 0.9. Under shorter deadlines H=1 delivers 87.49% versus 85.75% for the predictor. Simulation omits ACK loss, waveform effects, imperfect synchronization, multiple links, and real RF traces. Three seeds limit uncertainty estimates; H=1 and H=16 also differ in parameter count. Results apply to these modeled settings only.

6. Reproducibility and conclusion

The repository includes original source, tested dependencies, six portable checkpoints, hashed scenarios, per-episode results, training logs, tests, plots, and a fixed-scenario demo. CPU inference and exact same-platform continuation are tested. Conventional predictors remain serious alternatives; the contribution is a reproducible RL formulation and measured comparison under delayed, partially observed control. Codex assisted implementation; the repository records provenance.

References

[1] S. Wang, H. Liu, P. H. Gomes, and B. Krishnamachari. Deep Reinforcement Learning for Dynamic Multichannel Access in Wireless Networks. arXiv:1802.06958, 2018. https://arxiv.org/abs/1802.06958

[2] S. Ak and S. Bruggenwirth. Avoiding Jammers: A Reinforcement Learning Approach. arXiv:1911.08874, 2019. https://arxiv.org/abs/1911.08874

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[4] H. van Hasselt, A. Guez, and D. Silver. Deep Reinforcement Learning with Double Q-learning. arXiv:1509.06461, 2015. https://arxiv.org/abs/1509.06461